

Project Report: Real-Time Dashboard with WebSockets - Frontend React Developer

Written by Gagan Pasupuleti

Family:	Frontend React Developer
Level:	Advanced
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

React dashboard with live metrics via WebSocket, Recharts visualizations, auth, and error boundaries.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Advanced]

Chapter focus: Project Overview** *(Level: Advanced)

Real-Time Dashboard with WebSockets** is a advanced-level end-to-end project for Frontend React Developer. Duration: 4-5 weeks. React dashboard with live metrics via WebSocket, Recharts visualizations, auth, and error boundaries. Tools: React 19, Recharts, WebSocket, JWT.

Code Reference:

```
Objective: WebSocket live data feed
Objective: Recharts time-series charts
Objective: Auth-protected routes
Objective: Error boundaries + retry logic
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Frontend React Developer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Frontend React Developer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: WS Client [Advanced]

Chapter focus: Phase: WS Client** *(Level: Advanced)

Custom useWebSocket hook. Complete these tasks: Reconnect logic; Heartbeat. Deliverable for this phase: Hook unit tests.

Code Reference:

- [] Reconnect logic
- [] Heartbeat

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase WS Client is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Frontend React Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Hook unit tests to the teacher.

Chapter 3: Phase: Charts [Advanced]

Chapter focus: Phase: Charts** *(Level: Advanced)

Live updating Recharts. Complete these tasks: Buffer last 100 points; Pause/resume. Deliverable for this phase: Dashboard page.

Code Reference:

- [] Buffer last 100 points
- [] Pause/resume

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Charts is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Frontend React Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Dashboard page to the teacher.

Chapter 4: Phase: Auth [Advanced]**Chapter focus: Phase: Auth** *(Level: Advanced)**

JWT login flow. Complete these tasks: Protected routes; Token refresh. Deliverable for this phase: Login ? dashboard flow.

Code Reference:

- [] Protected routes
- [] Token refresh

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Auth is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Frontend React Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Login ? dashboard flow to the teacher.

Chapter 5: Phase: Harden [Advanced]**Chapter focus: Phase: Harden** *(Level: Advanced)**

Error boundaries, Suspense. Complete these tasks: Fallback UI; Sentry integration. Deliverable for this phase: Error handling demo video.

Code Reference:

- [] Fallback UI
- [] Sentry integration

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Harden is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Frontend React Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Error handling demo video to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Advanced]

Chapter focus: Final Deliverables and Review** *(Level: Advanced)

Submit all final deliverables: Live dashboard; Architecture doc; Error handling demo; Performance profile. Skills demonstrated: WebSockets, Charts, Auth. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: Live dashboard
Deliverable: Architecture doc
Deliverable: Error handling demo
Deliverable: Performance profile
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Live dashboard and passes the rubric.